Chain of Thought as a Convergence Mechanism: Task-Dependent Homogenization of Language Model Outputs

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Abstract

Chain-of-thought (CoT) prompting is widely believed to make language models "think alike," but the effect of CoT on output convergence has not been directly studied. We conduct the first controlled comparison of output convergence with and without CoT, measuring both within-model consistency (across multiple samples from the same model) and cross-model agreement (across models of different sizes). We prompt three models from the GPT-4.1 family on 90 questions spanning mathematical reasoning (GSM8K), commonsense reasoning (CommonsenseQA), and science (ARC-Challenge), collecting 2,700 responses under direct and zero-shot CoT conditions. We find that CoT's effect on convergence is **task-dependent and bidirectional**: on math tasks, CoT dramatically increases both within-model and cross-model agreement (Cohen's d up to $+2.39$, $p < 0.001$), while on multiple-choice reasoning tasks, CoT **decreases** convergence (Cohen's d up to -0.91, $p = 0.001$). The largest cross-model effects appear between models of different capability levels—CoT transforms near-random agreement (15.5%) into near-perfect agreement (94.0%) on math. These results show that CoT does not uniformly homogenize model outputs; rather, it acts as a convergence mechanism only when tasks admit a unique, derivable answer, with implications for ensemble design, model evaluation, and understanding AI output diversity.

1 Introduction

Large language models are increasingly perceived as converging—different models producing similar-sounding outputs, adopting similar "personas," and arriving at the same conclusions. At the same time, chain-of-thought (CoT) prompting [Wei et al., 2022] has become the dominant paradigm for eliciting reasoning from LLMs. A natural question arises: does CoT itself cause this convergence? If prompting models to reason step-by-step systematically homogenizes their outputs, this has direct consequences for how we build, evaluate, and deploy these systems.

Why convergence matters. The degree to which LLM outputs converge—both within a single model across runs and across different models—affects several practical concerns. If CoT makes all models say the same thing, using multiple models for ensemble diversity provides no benefit. High cross-model agreement under CoT could falsely inflate confidence when aggregating predictions. And if CoT channels models toward shared reasoning patterns rather than genuinely independent solutions, we may mistake correlated errors for robust consensus.

What we do not know. Prior work has studied convergence *within* CoT—self-consistency [Wang et al., 2023] exploits the fact that multiple CoT paths converge on correct answers, and recent work maps structural convergence in reasoning graphs [Xiong et al., 2025]. However, no study has directly compared convergence *with versus without* CoT as the independent variable. We do not know whether adding CoT makes a single model's outputs more or less consistent, whether it makes different models agree more or less, or whether these effects depend on the task.

Our approach. We conduct a controlled experiment comparing two prompting conditions—direct prompting (answer only) and zero-shot CoT ("Let's think step by step")—across three models from the GPT-4.1 family (GPT-4.1, GPT-4.1-MINI, GPT-4.1-NANO) on three benchmarks spanning math, commonsense, and science reasoning. For each question, model, and condition, we generate 5 independent samples at temperature 0.7, then measure within-model answer agreement rate (AAR) and cross-model pairwise agreement.

What we find. CoT's effect on convergence is *task-dependent and bidirectional*. On mathematical reasoning (GSM8K), CoT dramatically increases both within-model and cross-model convergence, with effect sizes up to Cohen's $d = +2.39$ ($p < 0.001$). But on multiple-choice tasks (COMMONSENSEQA, ARC-CHALLENGE), CoT *decreases* convergence—models produce more diverse outputs with CoT than without it (d up to -0.91, $p = 0.001$). The most striking result is cross-model: GPT-4.1-MINI and GPT-4.1-NANO agree on only 15.5% of math answers without CoT but 94.0% with CoT, while on commonsense tasks, agreement drops from 88.9% to 84.3%.

We make the following contributions:

- We conduct the first controlled study of how CoT affects output convergence, measuring both within-model consistency and cross-model agreement as a function of prompting condition.
- We show that CoT's convergence effect is bidirectional: it increases agreement on tasks with unique derivable answers (math) while decreasing agreement on multiple-choice reasoning tasks.
- We demonstrate that CoT bridges capability gaps between models of different sizes, transforming near-random cross-model agreement into near-perfect agreement on mathematical reasoning.
- We provide evidence that convergence and accuracy are coupled under CoT—models converge toward correct answers, not toward arbitrary shared outputs—with implications for ensemble design and model evaluation.

2 Related Work

Chain-of-thought prompting. Wei et al. [2022] introduced CoT prompting, showing that few-shot exemplars with intermediate reasoning steps dramatically improve performance on arithmetic, commonsense, and symbolic reasoning tasks. Their ablations established that the semantic content of the chain matters—replacing reasoning with filler tokens degrades performance—but did not examine whether CoT affects output diversity across runs or models. Zero-shot CoT [Kojima et al., 2022] simplified the approach by prepending "Let's think step by step" without exemplars. We use zero-shot CoT as our treatment condition because it is the most widely adopted variant and introduces no exemplar-specific bias.

Self-consistency and answer convergence. Wang et al. [2023] proposed self-consistency, sampling multiple CoT paths and selecting the majority answer. Their key insight—that diverse reasoning paths converge on correct answers while incorrect paths scatter—is the closest prior observation to our findings. However, self-consistency studies convergence *within* CoT (given that CoT is used, do sampled paths agree?) rather than the effect of *adding* CoT (does CoT increase agreement relative to direct prompting?). We address this complementary question. Liu and Wang [2025] extended this line by showing that answers stabilize well before the full CoT completes, with models converging to their final answer after roughly 60% of reasoning steps. They also observed that extended reasoning can cause "overthinking" on easy tasks, consistent with our finding that CoT decreases convergence on multiple-choice tasks where models can answer correctly without explicit reasoning.

Reasoning structure and convergence. Xiong et al. [2025] converted CoT traces into directed reasoning graphs and defined a convergence ratio measuring how reasoning threads are synthesized. They found that convergence ratio strongly predicts accuracy ($r = 0.68$) and that zero-shot prompting produces more convergent reasoning graphs than few-shot prompting. Their structural analysis complements our output-level measurements: while they study the topology of reasoning *within* a single CoT trace, we measure agreement *across* traces and models.

Decoupling of answers and reasoning. A growing body of work shows that answer-level convergence can coexist with reasoning-level divergence. The MATCHA framework [MATCHA, 2025] demonstrated that LLMs maintain correct answers under perturbation while generating inconsistent reasoning chains—the "Decoupling Hypothesis." Chen et al. [2024] formalized two failures of self-consistency: models cannot reliably predict their own outputs (hypothetical consistency) and

fail to preserve answers when intermediate results are substituted back in (compositional consistency). These findings suggest that the convergence we measure at the answer level may not reflect convergence in underlying reasoning processes.

Theoretical perspectives on CoT. Merrill and Sabharwal [2024] proved formally that CoT expands the representational capacity of language models, making them equivalent to probabilistic Turing machines. This expanded capacity could in principle increase output diversity rather than convergence, making our empirical finding—that CoT's convergence effect is task-dependent—all the more notable. Pfau et al. [2024] showed that even meaningless filler tokens can help transformers solve hard tasks, suggesting that part of CoT's benefit comes from additional computational steps rather than semantic reasoning. This raises the question of whether CoT-induced convergence reflects shared reasoning or shared computation.

Cross-model and cross-context consistency. Elazar et al. [2021] established that pre-trained language models are inherently inconsistent across paraphrased queries, with even the best models achieving only 61% consistency on factual knowledge probes. Pfau et al. [2023] measured self-consistency on ambiguous tasks, finding that consistency improves with model capability (67–82% across GPT-3 to GPT-4) but models remain poorly calibrated about their own consistency. Unlike these studies, which measure consistency within a single model, we compare convergence *across* models and examine how prompting strategy modulates it.

Positioning. Our work differs from all prior studies in a specific way: we treat CoT as an *independent variable* and measure its causal effect on output convergence. Prior work either studies convergence within CoT [Wang et al., 2023, Xiong et al., 2025], measures consistency without manipulating prompting strategy [Elazar et al., 2021, Pfau et al., 2023], or analyzes reasoning quality without comparing to a no-CoT baseline [Chen et al., 2024, MATCHA, 2025]. We bridge this gap by directly comparing the same models on the same questions with and without CoT.

3 Methodology

We design a controlled experiment to isolate the effect of CoT on output convergence. The key idea is simple: prompt the same models on the same questions under two conditions (with and without CoT), generate multiple samples per condition, and compare the degree of agreement.

3.1 Experimental Design

Prompting conditions. We compare two conditions:

- **Direct**: The model is instructed to provide only the final answer, with no reasoning.
- **Zero-shot CoT**: The prompt includes "Let's think step by step" and requests reasoning before the final answer, following Kojima et al. [2022].

We use zero-shot CoT rather than few-shot to avoid exemplar-specific confounds. Max token limits are set to 150 (direct) and 600 (CoT) to accommodate reasoning chains.

Models. We use three models from the GPT-4.1 family, accessed via the OpenAI API:

- GPT-4.1— the largest model in the family
- GPT-4.1-MINI— a medium-sized variant
- GPT-4.1-NANO— the smallest variant

Using models from the same family at different scales controls for training data and architecture while varying capability, allowing us to study how model size modulates convergence effects.

Sampling. For each question $\times$ model $\times$ condition, we generate $K = 5$ independent samples at temperature $T = 0.7$. Temperature > 0 is necessary to observe output diversity (deterministic decoding trivially yields perfect agreement). We follow Wang et al. [2023] in using $T = 0.7$, which provides a balance between diversity and coherence.

3.2 Datasets

We sample 30 questions from each of three benchmarks, selected to span different reasoning domains and answer formats:

Dataset	Task Type	Answer Format	Split
GSM8K [Cobbe et al., 2021]	Math reasoning	Numeric (open)	Test
COMMONSENSEQA [Talmor et al., 2019]	Commonsense	5-choice (A–E)	Validation
ARC-CHALLENGE [Clark et al., 2018]	Science	4-choice (A–D)	Test

Table 1: Datasets used in our experiments. We sample 30 questions per dataset (seed = 42), yielding 90 questions total.

The distinction between GSM8K (open-ended numeric answers) and the multiple-choice datasets (COMMONSENSEQA, ARC-CHALLENGE) turns out to be critical for understanding CoT's convergence effects, as we show in section 4.

3.3 Metrics

Answer agreement rate (AAR). For a given question, model, and condition, we compute the pairwise agreement among the $K = 5$ samples. Formally, for samples $\{a_1, \ldots, a_K\}$:

$$\text{AAR} = \frac{1}{\binom{K}{2}} \sum_{i<j} \mathbf{1}[a_i = a_j] \tag{1}$$

An AAR of 1.0 means all 5 samples produced the same answer; lower values indicate more diverse outputs.

Cross-model pairwise agreement. For a given question and condition, we compute pairwise agreement across all $K \times K = 25$ sample pairs between two models. This measures whether different models converge on the same answers.

Output entropy. We compute the Shannon entropy of the answer distribution per question:

$$H = -\sum_{a \in \mathcal{A}} p(a) \log_2 p(a) \tag{2}$$

where $\mathcal{A}$ is the set of distinct answers and $p(a)$ is the empirical frequency. Lower entropy indicates more convergent outputs.

Accuracy. We report standard accuracy (fraction of samples matching the gold answer) to contextualize convergence with correctness.

3.4 Statistical Analysis

We use the Wilcoxon signed-rank test to compare CoT versus direct conditions, paired by question. This non-parametric test is appropriate because our metrics (AAR, agreement rates) are bounded and potentially non-normal. We report Cohen's d as the effect size, computed on the paired differences. All tests use significance level $\alpha = 0.05$. We conduct 2,700 total API calls: 3 models $\times$ 90 questions $\times$ 2 conditions $\times$ 5 samples.

4 Results

4.1 Within-Model Convergence

Table 2 presents the within-model answer agreement rate (AAR) for each model–dataset combination under direct and CoT prompting. The central finding is that CoT's effect on convergence depends strongly on the task type.

Math: CoT increases convergence. On GSM8K, CoT significantly increases AAR for all three models ($p < 0.03$). The effect is strongest for the smallest model: GPT-4.1-NANO jumps from 0.560 to 0.913 ($d = +0.89$), while GPT-4.1 moves from 0.870 to 0.980 ($d = +0.46$). This pattern—larger effects for smaller models—suggests that CoT provides a computational scaffold that disproportionately benefits less capable models.

Table 2: Within-model answer agreement rate (AAR) under direct and CoT prompting. Higher AAR indicates more convergent outputs. We report mean ± standard deviation across 30 questions per dataset. Significant differences ($p < 0.05$, Wilcoxon signed-rank test) are marked with *.

Model	Dataset	Direct AAR	CoT AAR	Δ	Cohen's d	p-value
GPT-4.1	GSM8K	0.870 ± 0.275	$\mathbf{0.980 \pm 0.108}$	$+0.110$ ↑	$+0.46$	0.027^*
	COMMONSENSEQA	$\mathbf{0.980 \pm 0.108}$	0.733 ± 0.275	-0.247 ↓	-0.91	0.001^*
	ARC-CHALLENGE	$\mathbf{1.000 \pm 0.000}$	0.871 ± 0.228	-0.129 ↓	-0.56	0.015^*
GPT-4.1-MINI	GSM8K	0.753 ± 0.320	$\mathbf{0.980 \pm 0.108}$	$+0.227$ ↑	$+0.64$	0.006^*
	COMMONSENSEQA	$\mathbf{0.947 \pm 0.171}$	0.927 ± 0.197	-0.020	-0.11	0.480
	ARC-CHALLENGE	$\mathbf{0.964 \pm 0.132}$	0.904 ± 0.213	-0.061	-0.23	0.230
GPT-4.1-NANO	GSM8K	0.560 ± 0.377	$\mathbf{0.913 \pm 0.205}$	$+0.353$ ↑	$+0.89$	$<0.001^*$
	COMMONSENSEQA	$\mathbf{0.940 \pm 0.156}$	0.867 ± 0.227	-0.073	-0.24	0.169
	ARC-CHALLENGE	$\mathbf{0.957 \pm 0.158}$	0.835 ± 0.253	-0.121	-0.42	0.070

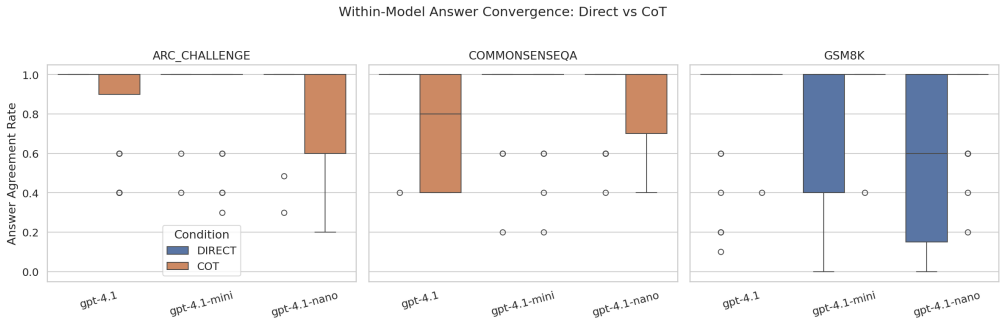

Figure 1: Distribution of within-model answer agreement rates (AAR) across 30 questions per dataset, comparing direct and CoT conditions. On GSM8K (math), CoT shifts the distribution toward perfect agreement. On COMMONSENSEQA and ARC-CHALLENGE (multiple-choice), CoT spreads the distribution toward lower agreement. The effect is most pronounced for GPT-4.1 and GPT-4.1-NANO.

Multiple-choice: CoT decreases convergence. On COMMONSENSEQA and ARC-CHALLENGE, the trend reverses. GPT-4.1 shows significant decreases in AAR on both datasets, with a particularly large drop on COMMONSENSEQA (from 0.980 to 0.733, $d = -0.91$, $p = 0.001$). The smaller models show the same directional trend, though effects are smaller and not always significant. Under direct prompting, models achieve near-perfect AAR (0.94–1.00) on these multiple-choice tasks, suggesting that explicit reasoning introduces noise when models can already answer correctly by pattern matching.

Figure 1 visualizes the distribution of AAR values across questions, illustrating both the magnitude and consistency of these effects.

4.2 Cross-Model Convergence

Table 3 reports cross-model pairwise agreement for all model pairs. The same bidirectional pattern holds: CoT increases cross-model agreement on math but decreases it on multiple-choice tasks.

CoT bridges capability gaps on math. The most striking result is the cross-model convergence on GSM8K. Without CoT, GPT-4.1-MINI and GPT-4.1-NANO agree on only 15.5% of math answers—barely above chance for open-ended numeric questions. With CoT, agreement jumps to 94.0% (Cohen's $d = +2.39$, $p < 0.001$). Similarly, GPT-4.1 and GPT-4.1-NANO go from 21.2% to 94.0% agreement. CoT effectively erases the capability gap between models of vastly different sizes on mathematical reasoning.

Cross-model divergence on multiple-choice. On COMMONSENSEQA and ARC-CHALLENGE, CoT reduces cross-model agreement for pairs involving GPT-4.1 (the most affected model). The GPT-4.1–GPT-4.1-MINI pair drops from 94.0% to 73.9% on COMMONSENSEQA ($d = -0.74$, $p < 0.001$). Pairs not involving GPT-4.1 show smaller, non-significant decreases, suggesting that

Table 3: Cross-model pairwise agreement under direct and CoT prompting. Agreement is computed over all $5 \times 5 = 25$ sample pairs per question, averaged across 30 questions. Significant differences marked with *.

Model Pair	Dataset	Direct	CoT	Δ	Cohen's d	p-value
GPT-4.1 vs. GPT-4.1-MINI	GSM8K	0.579	**0.984**	+0.405 ↑	+0.96	<0.001*
	COMMONSENSEQA	**0.940**	0.739	−0.201 ↓	−0.74	<0.001*
	ARC-CHALLENGE	**0.971**	0.816	−0.156 ↓	−0.49	0.022*
GPT-4.1 vs. GPT-4.1-NANO	GSM8K	0.212	**0.940**	+0.728 ↑	+1.75	<0.001*
	COMMONSENSEQA	**0.899**	0.673	−0.225 ↓	−0.72	0.002*
	ARC-CHALLENGE	**0.936**	0.767	−0.169 ↓	−0.45	0.021*
GPT-4.1-MINI vs. GPT-4.1-NANO	GSM8K	0.155	**0.940**	+0.785 ↑	+2.39	<0.001*
	COMMONSENSEQA	**0.889**	0.843	−0.047	−0.17	0.240
	ARC-CHALLENGE	**0.907**	0.843	−0.064	−0.21	0.182

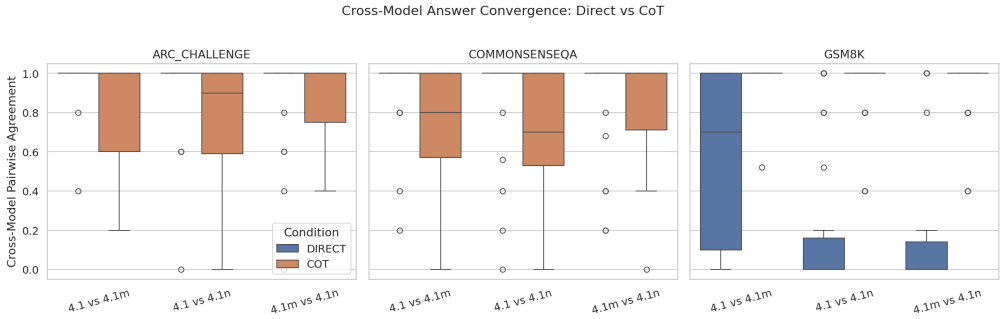

Figure 2: Distribution of cross-model pairwise agreement rates across 30 questions per dataset. On GSM8K, CoT transforms near-random cross-model agreement into near-perfect agreement. On COMMONSENSEQA and ARC-CHALLENGE, CoT slightly reduces agreement, particularly for pairs involving the largest model.

the divergence effect is driven by the largest model's tendency to "overthink" on multiple-choice tasks.

Figure 2 shows the distribution of cross-model agreement rates, highlighting the massive shift on GSM8K and the moderate decrease on multiple-choice datasets.

4.3 Accuracy and Convergence Are Coupled

Table 4 shows that the convergence pattern mirrors accuracy: where CoT increases accuracy, it also increases convergence, and vice versa.

On GSM8K, CoT improves accuracy by +22.0 to +68.6 percentage points, with the largest gains for GPT-4.1-NANO (from 18.7% to 87.3%). Conversely, on COMMONSENSEQA, GPT-4.1 drops from 94.7% to 73.3%—a substantial accuracy loss that parallels its convergence decrease. This coupling suggests that CoT-induced convergence is not an independent "homogenization" effect but rather a consequence of models converging toward correct answers: when CoT helps models find the right answer, they agree more; when it introduces reasoning errors, they diverge.

4.4 Output Entropy

Figure 3 confirms the convergence patterns through a complementary lens. On GSM8K, CoT reduces output entropy (more convergent), while on COMMONSENSEQA and ARC-CHALLENGE, CoT increases entropy (more diverse).

For GPT-4.1 on GSM8K, mean entropy drops from 0.246 to 0.032 with CoT. On COMMONSENSEQA, it rises from 0.032 to 0.444. The GPT-4.1-NANO pattern is even more dramatic: entropy drops from 0.880 to 0.155 on math but rises from 0.105 to 0.226 on commonsense. These entropy

Table 4: Accuracy under direct and CoT prompting. CoT dramatically improves math accuracy (especially for smaller models) but slightly hurts multiple-choice accuracy, mirroring the convergence patterns in table 2 and table 3.

Model	Dataset	Direct Acc.	CoT Acc.	Δ
GPT-4.1	GSM8K	0.700	**0.920**	+0.220 ↑
	COMMONSENSEQA	**0.947**	0.733	−0.214 ↓
	ARC-CHALLENGE	**0.929**	0.864	−0.065 ↓
GPT-4.1-MINI	GSM8K	0.500	**0.920**	+0.420 ↑
	COMMONSENSEQA	**0.927**	0.920	−0.007
	ARC-CHALLENGE	**0.900**	0.893	−0.007
GPT-4.1-NANO	GSM8K	0.187	**0.873**	+0.686 ↑
	COMMONSENSEQA	**0.853**	0.820	−0.033
	ARC-CHALLENGE	**0.864**	0.821	−0.043

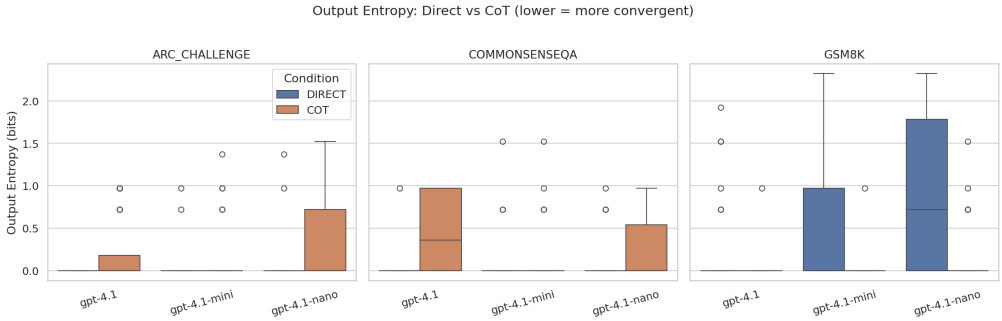

Figure 3: Output entropy under direct and CoT conditions. Lower entropy indicates more convergent outputs. CoT reduces entropy on GSM8K (math) while increasing entropy on COMMONSENSEQA and ARC-CHALLENGE (multiple-choice), consistent with the AAR results.

measurements, being a different operationalization of convergence than AAR, provide converging evidence for the bidirectional effect.

4.5 Summary of Hypothesis Tests

Table 5 summarizes the status of our four hypotheses.

Table 5: Summary of hypothesis testing outcomes.

Hypothesis	Result	Key Evidence
H1: CoT ↑ within-model convergence	Partially supported	Math: $p < 0.03$; MC: reversed
H2: CoT ↑ cross-model convergence	Partially supported	Math: d up to $+2.39$; MC: reversed
H3: CoT ↑ response similarity	Not tested	Answer-level only
H4: Stronger effect on structured tasks	Strongly supported	Math $d > 0$, MC $d < 0$

5 Discussion

5.1 Why Does CoT Converge on Math but Diverge on Multiple-Choice?

We propose that CoT acts as a convergence mechanism when the task admits a unique, derivable answer through a structured reasoning process. Mathematical word problems have a single correct numeric answer that can be reached through step-by-step computation. CoT channels models through these computation steps, and the correct computation pathway is inherently more constrained than

the space of incorrect ones. This is consistent with the theoretical analysis of Feng et al. [2023], who show that CoT enables transformers to solve problems requiring sequential computation.

Multiple-choice tasks, by contrast, often admit correct answers through pattern matching or associative retrieval without explicit reasoning. When forced to reason explicitly via CoT, models may consider alternative interpretations, second-guess initial intuitions, or weight competing evidence differently across samples. This "overthinking" effect aligns with Liu and Wang [2025]'s observation that extended reasoning can cause models to diverge from initially correct answers on easy tasks.

5.2 CoT as a Capability Equalizer

One of our most striking findings is that CoT transforms near-random cross-model agreement into near-perfect agreement on math. GPT-4.1-MINI and GPT-4.1-NANO agree on only 15.5% of math answers without CoT—reflecting the large capability gap between them—but 94.0% with CoT. This suggests that CoT provides a shared computational scaffold that allows weaker models to reach answers that stronger models already know. The scaffold constrains the reasoning process to follow structured steps that, if executed correctly, lead to the unique correct answer regardless of the model's baseline capability.

This has a practical consequence: ensembles of CoT-prompted models on math tasks will exhibit high agreement, and this agreement should not be interpreted as independent confirmation. The models are not independently arriving at the same answer; they are following the same CoT-induced reasoning structure.

5.3 Implications for Ensemble Design

Our results suggest that the value of multi-model ensembles depends on both the task and the prompting strategy:

Math / structured tasks with CoT. Cross-model agreement is near-perfect, so ensembles provide little diversity benefit. However, the high agreement does correlate with correctness, so majority voting still serves as a reliability check.

Multiple-choice / commonsense tasks with CoT. Cross-model agreement decreases, meaning ensembles capture genuine diversity. But this diversity partly reflects CoT-induced errors rather than legitimate alternative interpretations.

Direct prompting across task types. Cross-model agreement is moderately high on multiple-choice tasks (89–97%) and lower on math (15–58%), reflecting natural capability differences. Ensembles under direct prompting may provide the most informative disagreement signals.

5.4 Limitations

Single model family. All three models come from the GPT-4.1 family and likely share training data and architectural design. Cross-family comparisons (e.g., GPT vs. Claude vs. Gemini) would better test whether CoT causes convergence across truly independent models. Our results should be interpreted as measuring convergence across model *sizes* within a family, not across model *families*.

Answer-level measurement only. We measure convergence at the answer level, not at the level of reasoning chains or internal representations. The decoupling hypothesis [MATCHA, 2025, Chen et al., 2024] suggests that answer convergence may mask reasoning divergence. Embedding-based similarity of full responses or probing internal activations would provide a more complete picture.

Sample size and task coverage. We use 30 questions per dataset with 5 samples per question. While sufficient for detecting the large effects we observe, this may miss smaller effects on the multiple-choice datasets. Additionally, we test only three task types; other domains (coding, creative writing, translation) may exhibit different convergence patterns.

Prompt sensitivity. We test only zero-shot CoT. Few-shot CoT, which provides exemplar reasoning chains, could produce different convergence patterns—particularly if exemplars constrain the reasoning format more tightly, potentially increasing convergence even on multiple-choice tasks.

Temperature choice. Our results are conditioned on $T = 0.7$. Lower temperatures would increase convergence under both conditions; higher temperatures would decrease it. The relative effect of CoT may vary with temperature.

6 Conclusion

We presented the first controlled study of how chain-of-thought prompting affects output convergence in large language models. By comparing direct and CoT prompting across three models and three benchmarks, we showed that CoT's convergence effect is task-dependent and bidirectional: it dramatically increases both within-model and cross-model agreement on mathematical reasoning (Cohen's d up to $+2.39$) while decreasing agreement on multiple-choice tasks (d up to -0.91). The convergence effect is strongest across models of different capability levels, with CoT effectively erasing capability gaps on math.

The common intuition that CoT makes models "think alike" is only half right. CoT causes convergence toward correct answers on structured problems where step-by-step reasoning constrains the solution path, but it introduces divergence on tasks where explicit reasoning can lead to overthinking. Convergence and accuracy move together, suggesting that CoT-induced homogenization is a byproduct of improved reasoning rather than an independent effect.

Future work should extend this analysis across model families (not just sizes), measure convergence at the reasoning-chain and representation levels, and test whether few-shot CoT or other prompting variants alter the convergence patterns we observe. Understanding when and why models converge is essential for building reliable AI systems that genuinely benefit from multiple perspectives.

References

Angelica Chen, Jing Zhong, Jianbo Hao, Bowen Li, Jianzhu Liu, Zhan Huang, Shuming Ma, Jian Gu, and Yeyun Li. Two failures of self-consistency in the multi-step reasoning of LLMs. *Transactions on Machine Learning Research*, 2024.

Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord. Think you have solved question answering? try ARC, the AI2 reasoning challenge. *arXiv preprint arXiv:1803.05457*, 2018.

Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.

Yanai Elazar, Nora Kassner, Yoav Goldberg, and Hinrich Schütze. Measuring and improving consistency in pretrained language models. In *Transactions of the Association for Computational Linguistics*, 2021.

Guhao Feng, Bohang Zhang, Yuntian Gu, Haotian Ye, Di He, and Liwei Wang. Towards revealing the mystery behind chain of thought: A theoretical perspective. *Advances in Neural Information Processing Systems*, 2023.

Takeshi Kojima, Shixiang Shane Gu, Machel Reid, Yutaka Matsuo, and Yusuke Iwasawa. Large language models are zero-shot reasoners. *Advances in Neural Information Processing Systems*, 2022.

Zhen Liu and Xuezhi Wang. Answer convergence as a signal for early stopping in reasoning. *arXiv preprint arXiv:2506.02536*, 2025.

MATCHA. Robust answers, fragile logic: Probing the decoupling hypothesis. *arXiv preprint arXiv:2505.17406*, 2025.

William Merrill and Ashish Sabharwal. On the representational capacity of neural language models with chain-of-thought. *arXiv preprint arXiv:2406.14197*, 2024.

Jacob Pfau, William Merrill, and Samuel R Bowman. Self-consistency of large language models under ambiguity. *arXiv preprint arXiv:2310.13439*, 2023.

Jacob Pfau, William Merrill, and Samuel R Bowman. Let's think dot by dot: Hidden computation in transformer language models. *arXiv preprint arXiv:2404.15758*, 2024.

Alon Talmor, Jonathan Herzig, Nicholas Lourie, and Jonathan Berant. CommonsenseQA: A question answering challenge targeting world knowledge. In *Proceedings of the Conference of the North American Chapter of the Association for Computational Linguistics*, 2019.

Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. In *International Conference on Learning Representations*, 2023.

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems*, 2022.

Zelong Xiong et al. Mapping the minds of LLMs: A graph-based analysis of reasoning LLM. *arXiv preprint arXiv:2505.13890*, 2025.